



Wrapped one-class Riemannian EEG classifiers for BCI-based detection of anesthetic states

V. Marissens Cueva^{1,2*}, T. de Surrel^{3*}, L. Bougrain^{2,4}, S.J. Bidgoli⁵, G. Cheron⁶, A.M. Cebolla Alvarez⁶, C. Meistelman⁷, F. Lotte¹, F. Yger⁸, S. Rimbart¹

*These authors contributed equally to the paper

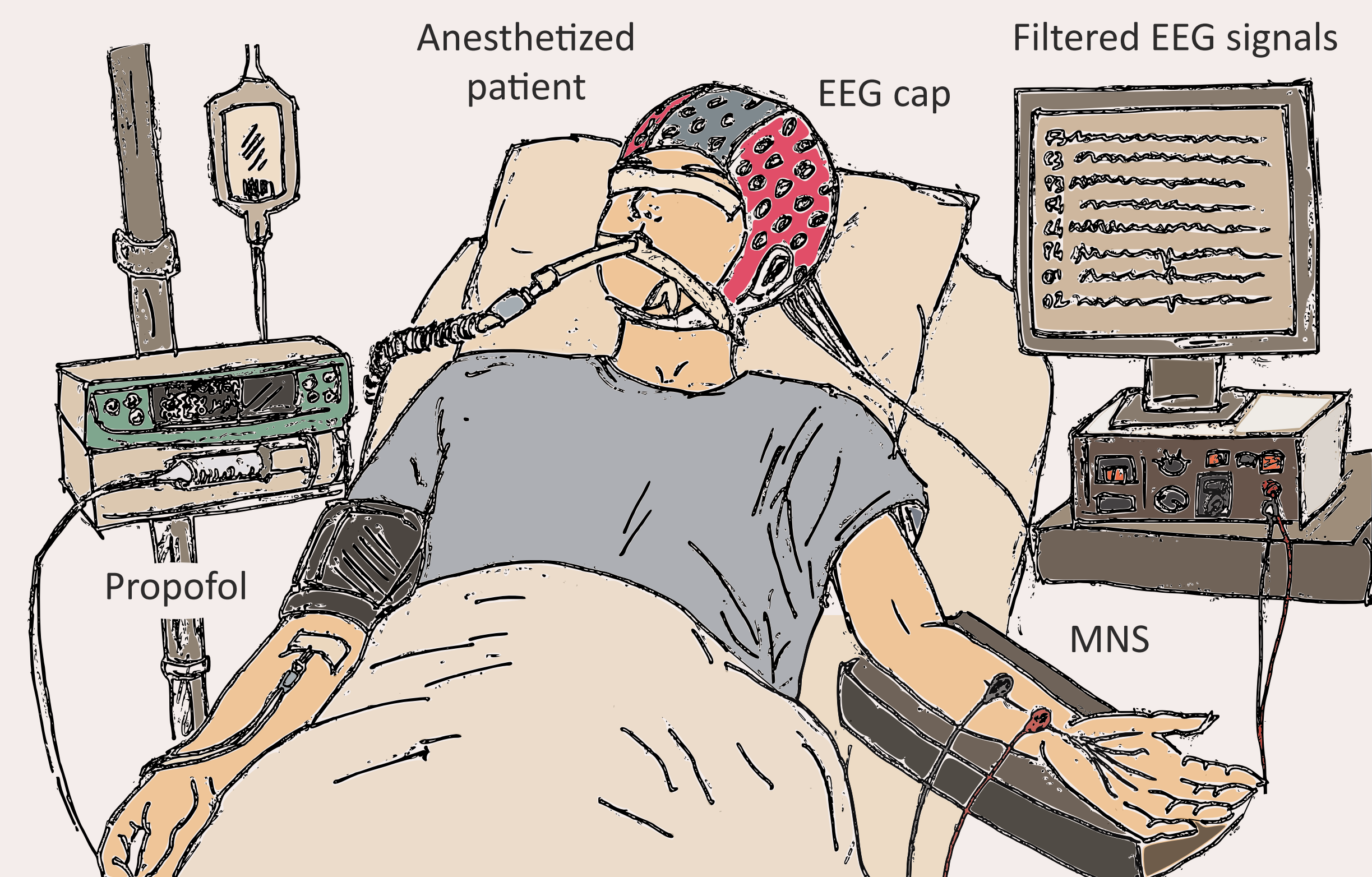
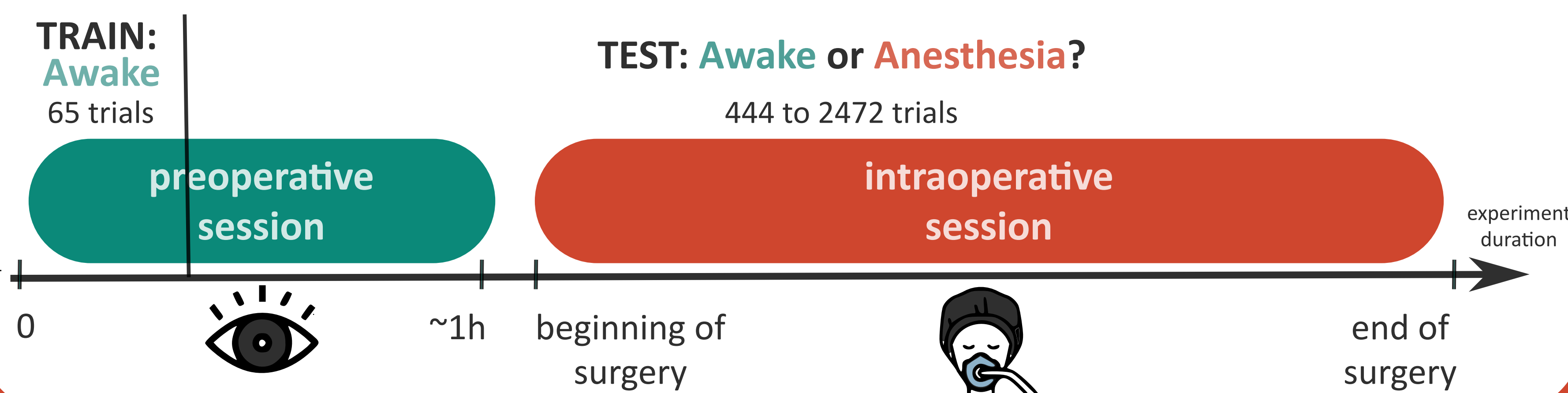
¹Inria Center at Univ. Bordeaux / LaBRI, Talence, France; ²Université de Lorraine, CNRS, LORIA, Nancy, France; ³LAMSADE, CNRS, PSL Univ. Paris-Dauphine, France; ⁴Sorbonne Université, ICM, CNRS, Inria, Inserm, Paris, France; ⁵CHU Brugmann, Bruxelles, Belgium; ⁶Laboratory of Neurophysiology and Movement Biomechanics, Université Libre de Bruxelles, Bruxelles, Belgium; ⁷DevAH, University of Lorraine, Nancy, France; ⁸LITIS, INSA Rouen-Normandie, France;

OBJECTIVE

Distinguish **awake** from **anesthetized** states with EEG, when only **awake** class is available, with a **new one-class algorithm**.

1 CLINICAL MOTIVATION & EXPERIMENT

Current brain monitors for detecting **intraoperative awareness** remain debated [Laferrière-Langois et al, *Anesth Analg*, 2024]. We propose a **Median Nerve Stimulation (MNS)-based BCI** to detect deviations from the awake state during anesthesia [Rimbart et al, *Front Neurosci*, 2019]. **19 patients** (10 women; 47±12.51 years) underwent surgery under general anesthesia with **propofol** at CHU Brugmann (Belgium), with EEG recorded in two sessions and MNS delivered every 3-4 s:



Drawing inspired by Sigl and Chamoun, *J Clin Monit*, 1994 and Hendry et al, *Neuromod*, 2024

2 ONE-CLASS ALGORITHMS

Using the covariance matrices of Median Nerve Stimulation-evoked EEG signals:

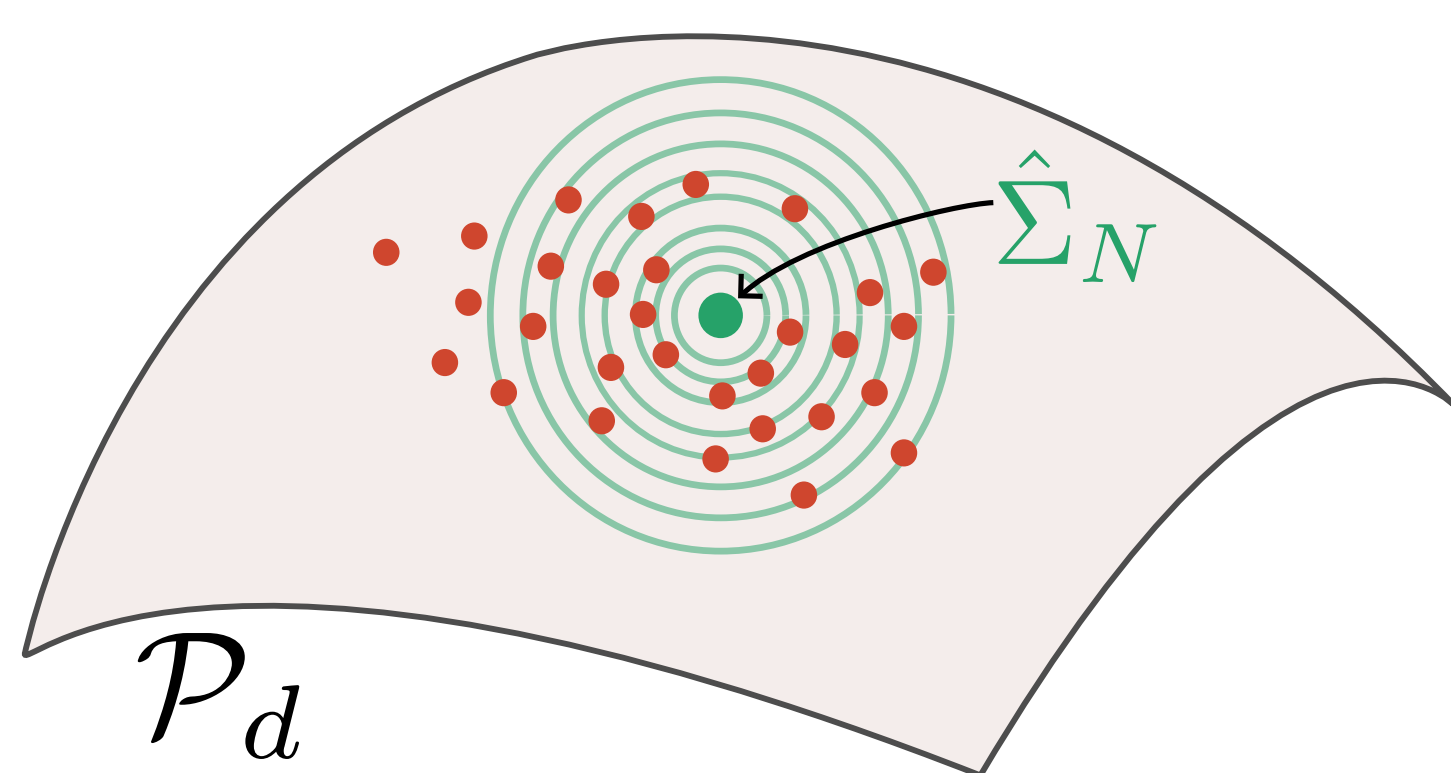
ONE-CLASS MINIMUM DISTANCE TO THE MEAN

Estimate: Riemannian mean $\hat{\Sigma}_N$

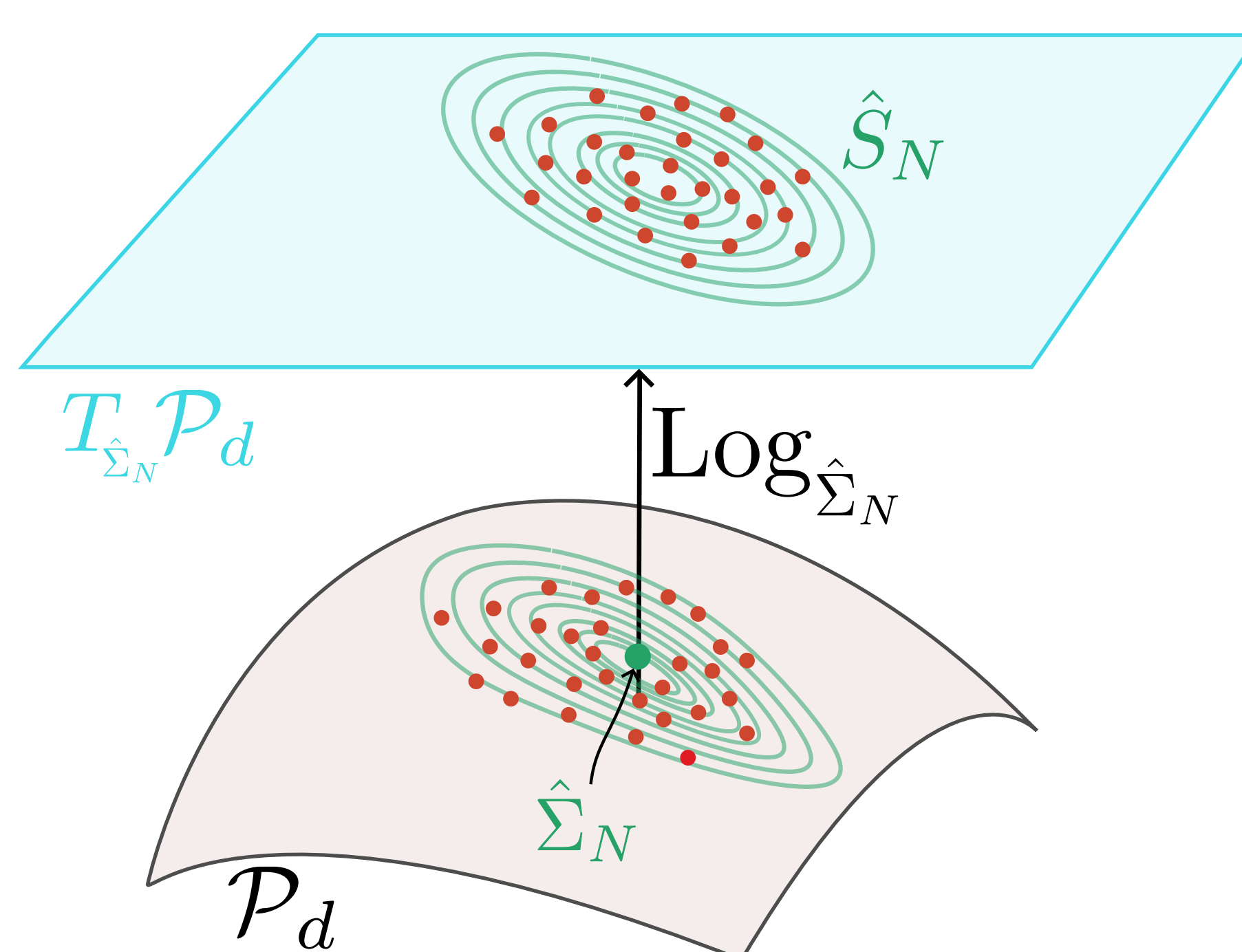
Classification rule:

$$\hat{y}_Z = \begin{cases} \text{awake} & \text{if } \delta(Z, \hat{\Sigma}_N) \leq \varepsilon, \\ \text{anesthetized} & \text{if } \delta(Z, \hat{\Sigma}_N) > \varepsilon, \end{cases}$$

with δ the Riemannian distance



ONE-CLASS WRAPPED GAUSSIAN (PROPOSED)



Estimate: Riemannian mean $\hat{\Sigma}_N$ and dispersion matrix $\hat{S}_N$

Classification rule:

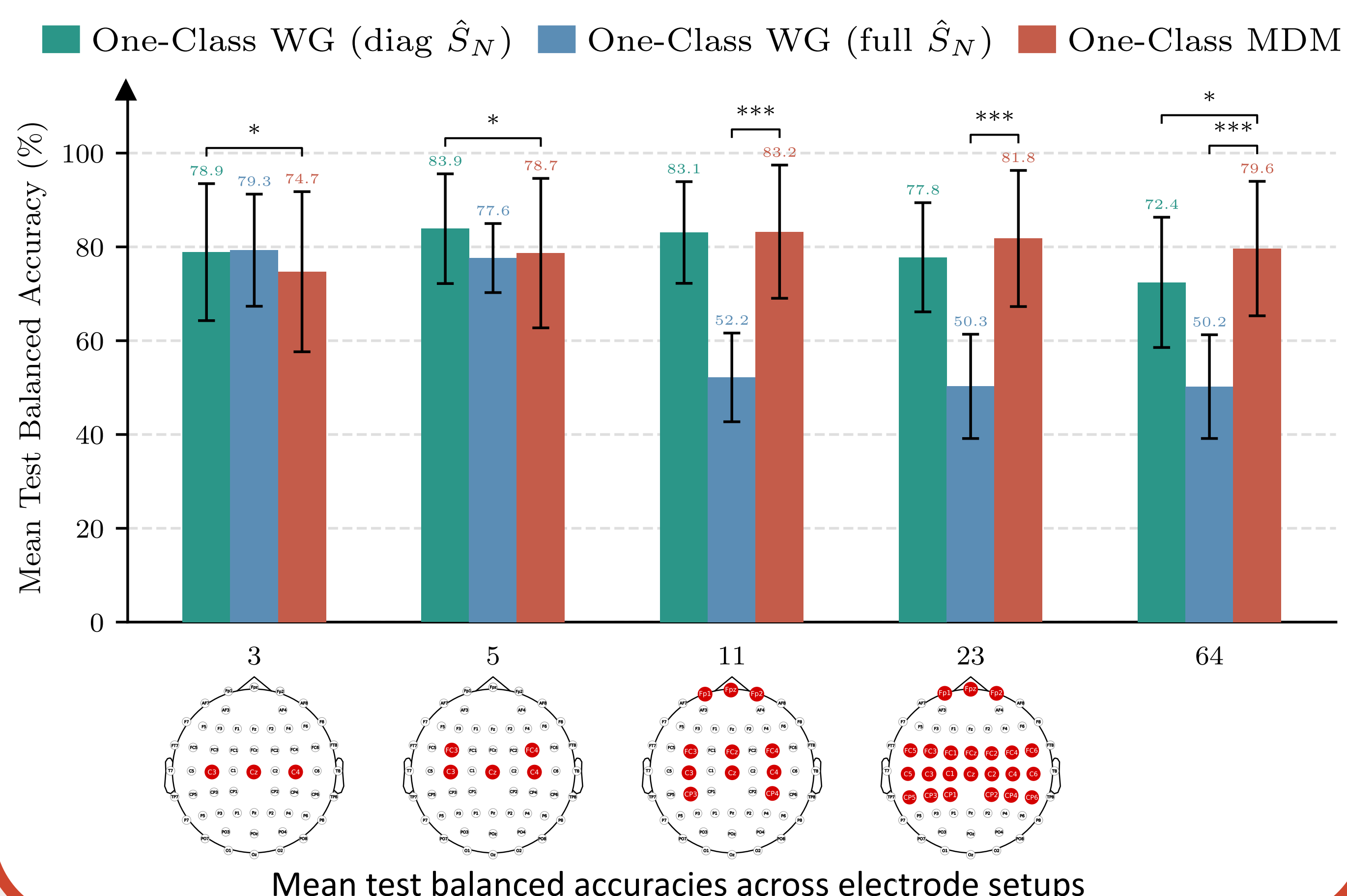
$$\hat{y}_Z = \begin{cases} \text{awake} & \text{if } \ell(Z) \geq \varepsilon, \\ \text{anesthetized} & \text{if } \ell(Z) < \varepsilon, \end{cases}$$

with ℓ the likelihood of the wrapped Gaussian

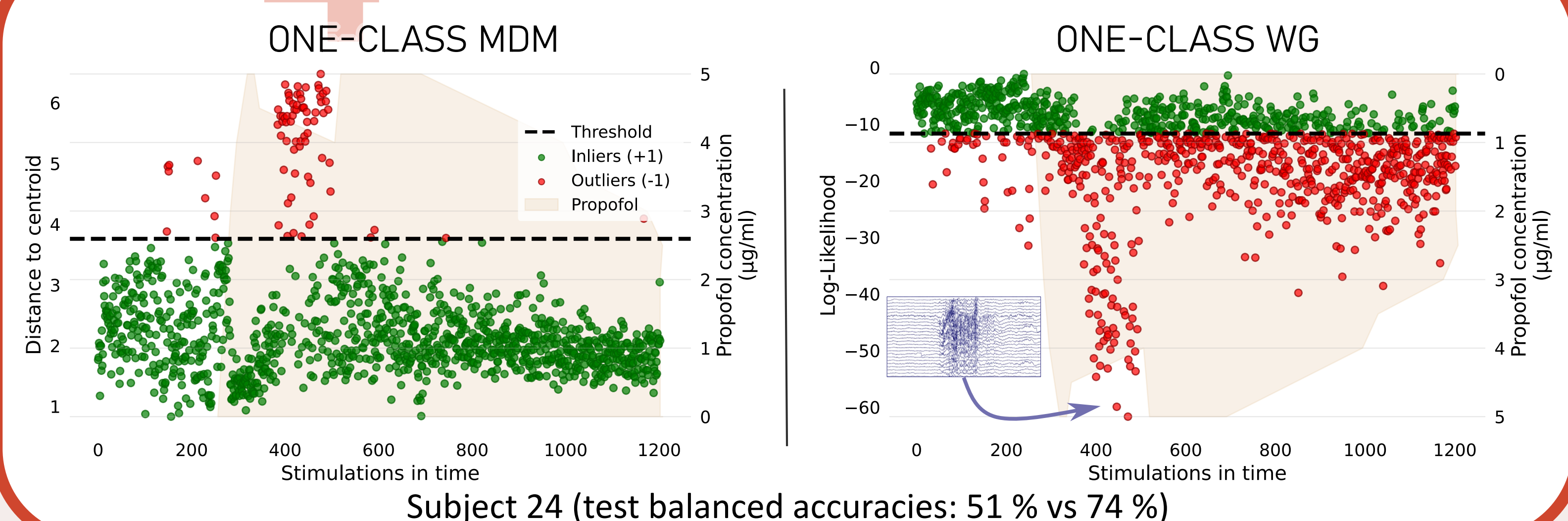
The **One-Class WG** is based on the **wrapped Gaussian** distribution [de Surrel, *Proc. ICML*, 2025], a Gaussian distribution defined on the tangent space and then send on the manifold.

Main difference: the One-Class MDM is **isotropic** while the One-Class WG is **anisotropic**.

3 AWAKE VS ANESTHESIA



4 COMPARISON WITH PROPOFOL



5 TAKE-HOME MESSAGE

Learning **only the awake EEG state** is **sufficient** to detect deviations associated with anesthesia, using a **reduced EEG montage**, with an **anisotropic Riemannian One-Class model** based on wrapped Gaussian

Acknowledgments: We thank the ANR BCI4IA & PROTEUS projects.